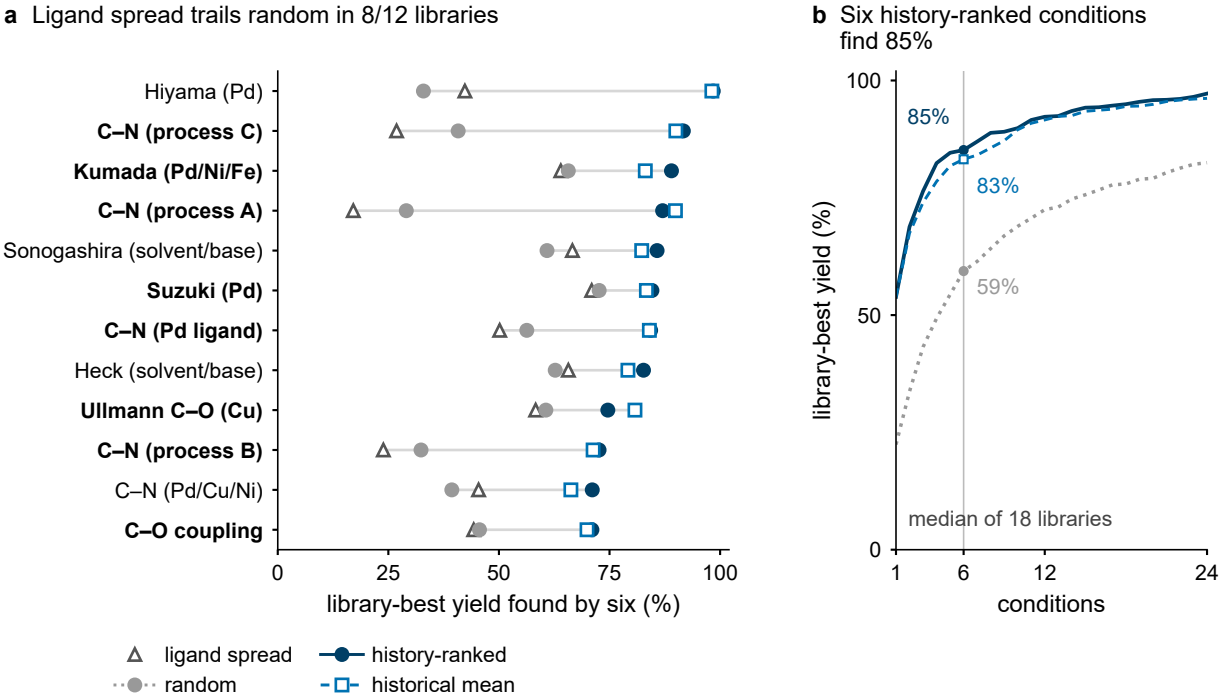
a Ligand spread trails random in 8/12 libraries
Hiyama (Pd)
C–N (process C)
Kumada (Pd/Ni/Fe)
C–N (process A)
Sonogashira (solvent/base)
Suzuki (Pd)
C–N (Pd ligand)
Heck (solvent/base)
Ullmann C–O (Cu)
C–N (process B)
C–N (Pd/Cu/Ni)
C–O coupling
0
25
50
75
100
library-best yield found by six (%)
ligand spread
random
history-ranked
historical mean
b Six history-ranked conditions find 85%
100
50
0
library-best yield (%)
85%
83%
59%
median of 18 libraries
1
6
12
24
conditions